

Problems Roster

Problem 1: Design a 3-bit Flash ADC for a signal with maximum & minimum value of 10 V & 0 V respectively. The quantized levels of the ADC should be uniform (equal step-size). Answer the following:

(a)	Calculate the total number of resistors & Op Amps required.
(b)	Calculate the 1LSB value or step size.
(c)	Find the quantization levels & plot the D_{out} (digital output) vs V_{in} plot.
(d)	If the input is 7 V: (i) Find the quantization interval the input lies in. (ii) What is the digital output for this input?
(e)	If the digital output is 110_2 , what are the maximum & minimum values of input that would produce this result?
(f)	If an additional bit was desired at the output, what would be the required number of components?

Solution:

(a) $N_R = 2^n = 2^3 = 8$

$N_{OA} = 2^n$ or $2^n - 1 = 8$ or 7

(b) $V_{1LSB} = \Delta = \frac{V_{max} - V_{min}}{2^n} = \frac{10 - 0}{8} = 1.25 \text{ V}$

(c) The quantization levels are as follows:

Quantization Interval	Quantization Level
0-1.25	0.625
1.25-2.5	1.875
2.5-3.75	3.125
3.75-5	4.375
5-6.25	5.625
6.25-7.5	6.875
7.5-8.75	8.125
8.75-10	9.375

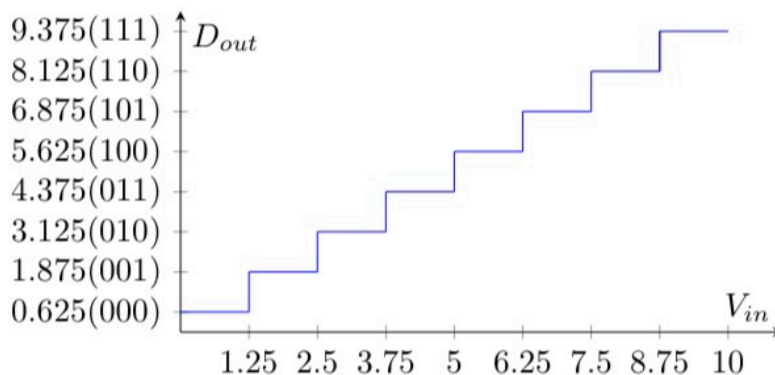
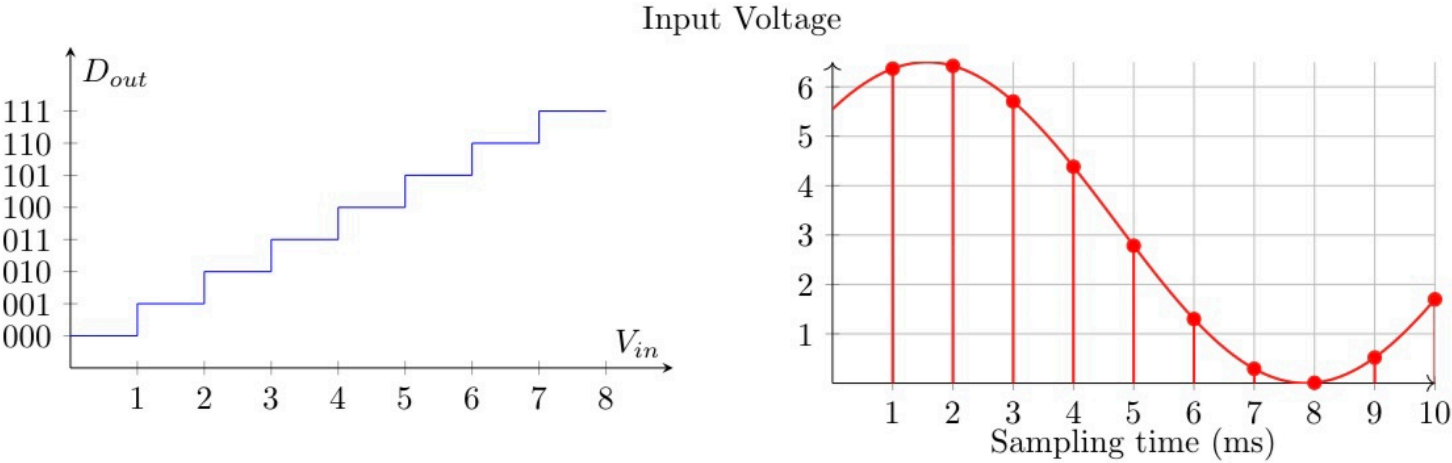


Figure: D_{out} vs V_{in} graph

- (d)
- (i) 6.25-7.5
- (ii) 101
- (e) Maximum = 8.75 V & minimum = 7.5 V.
- (f) For 4-bit Flash ADC, $N_{OPAMP} = 2^4$ or $2^4 - 1$. $N_R = 2^4$.

Problem 2:



(a)	Design a 3-bit Flash ADC for the given input-output characteristics.
(b)	An unknown signal is passed to the ADC as input. The ADC takes samples at a rate of 1 kHz. In the graph, the input signal is shown where the x-axis represents the time, and the y-axis represents voltage at any specific time. The sampling instances are given in the figure. Find the encoded output that represents the analog signal inside the given time frame.

Solution:

- (a)

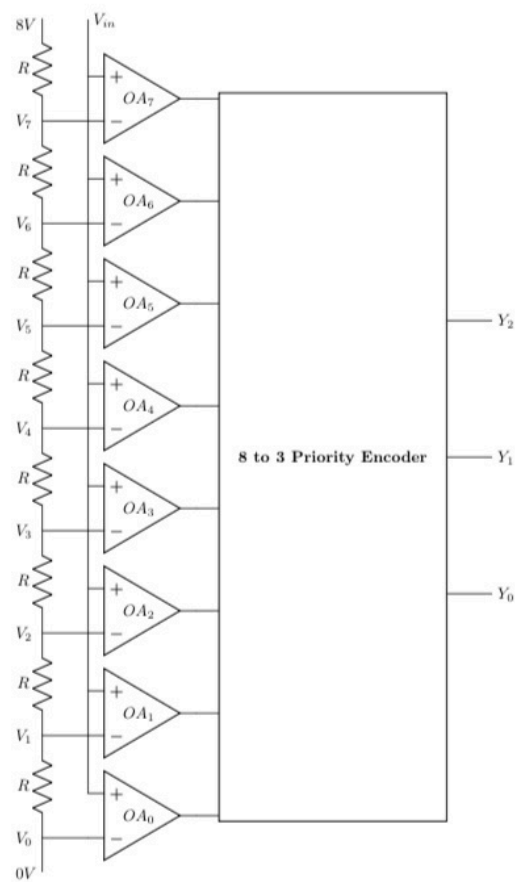


Figure 1.12: 3-bit Flash ADC for Problem 2(a)

(b)

Sampling Instances	Input Voltage	Quantization Interval	Encoded Value
1 ms	6.3 V	6 V - 7 V	110
2 ms	6.4 V	6 V - 7 V	110
3 ms	5.7 V	5 V - 6 V	101
4 ms	4.4 V	4 V - 5 V	100
5 ms	2.8 V	2 V - 3 V	010
6 ms	1.3 V	1 V - 2 V	001
7 ms	0.3 V	0 V - 1 V	000
8 ms	0 V	0 V - 1 V	000
9 ms	0.5 V	0 V - 1 V	000
10 ms	1.7 V	1 V - 2 V	001

Problem 3:

A Dual Slope ADC has $V_{ref} = 5 \text{ V}$, and an 8 bit counter that outputs an 8-bit representation of the input signal. A 1 MHz clock is used for the clock and counter circuits.

(a)	Determine the input voltage range, number of steps and step size (resolution) of the ADC.
(b)	For a specific input voltage, the ADC outputs 100. Determine the input voltage.
(c)	Determine the total time required to get the reading in (b).
(d)	Calculate the maximum sampling rate of the ADC.
(e)	A person wishes to get a sampling rate 4 times that of the current circuit. Determine how many bits should be used for the counter without changing any other system parameters.

Solution:

(a) Input voltage range: 0 V - 5 V

Number of steps = $2^8 = 256$

Step size, $\Delta = \frac{V_{ref}}{2^n} = \frac{5}{256} = 0.0195 \text{ V}$

(b) We know from equation 1.6, $V_{in} = \frac{N+1}{2^n} V_{ref}$

$$\therefore V_{in} = \frac{100 + 1}{256} \times 5 = 1.973 \text{ V}$$

(c) We know from equation 1.4, $t_1 = \frac{2^n}{f_{CLK}}$

$$\therefore t_1 = \frac{2^8}{1 \times 10^6} \text{ s} = 0.256 \text{ ms}$$

From equation 1.5, $t_2 = \frac{2^n}{f_{CLK}} \cdot \frac{V_{in}}{V_{ref}}$

$$\therefore t_2 = \frac{2^8}{1 \times 10^6} \cdot \frac{1.973}{5} \text{ s} = 0.101 \text{ ms}$$

Therefore, total conversion time, $t = t_1 + t_2 = 0.256 + 0.101 = 0.357 \text{ ms}$

(d) Maximum conversion time is limited by the time required for worst case conversion of input = $V_{ref} = 5 \text{ V}$. For this case $t_1 = t_2 = 0.256 \text{ ms}$.

Thus, worst case conversion time, $t_{max} = 2 \times t_1 = 2 \times 0.256 = 0.512 \text{ ms}$.

$$\therefore \text{Maximum sampling rate} = \frac{1}{t_{max}} = \frac{1}{0.512 \times 10^{-3}} \text{ Hz} = 1953.125 \text{ Hz}$$

(e) From (d) we see, maximum sampling rate $\propto \frac{1}{2^n}$. Thus,

$$\frac{f_{samp1}}{f_{samp2}} = \frac{2^{n_2}}{2^{n_1}} \Rightarrow \frac{2^{n_2}}{2^8} = \frac{1}{4} \Rightarrow n_2 = 6$$

Problem 4:

Suppose a sinusoidal signal with peak-to-peak value of 6 V & frequency of 1 kHz needs to be encoded using Dual Slope ADC. What changes are necessary to the Dual Slope ADC in *Problem*

3 to get a successful conversion?

Solution: If we closely observe, there is another limitation of the Dual Slope ADC of figure 1.7—it cannot deal with bipolar input signals (i.e. signals with both positive & negative values).

The sinusoidal signal mentioned in the question ranges between -3 to 3. To handle this signal and to get a successful conversion using the Dual Slope ADC, we first need to make all of its values positive. This can be done by adding an *appropriate offset* to the sine wave. The goal is to get all of its values within the range of 0 to a maximum value, which is the requirement for our Dual Slope ADC.

So, for a sine wave of amplitude 3 V (peak-to-peak value of 6 V), we need to add 3 V of offset. This will lift all the values of the sine wave between 0 V to 6 V. This is illustrated in the following figure:

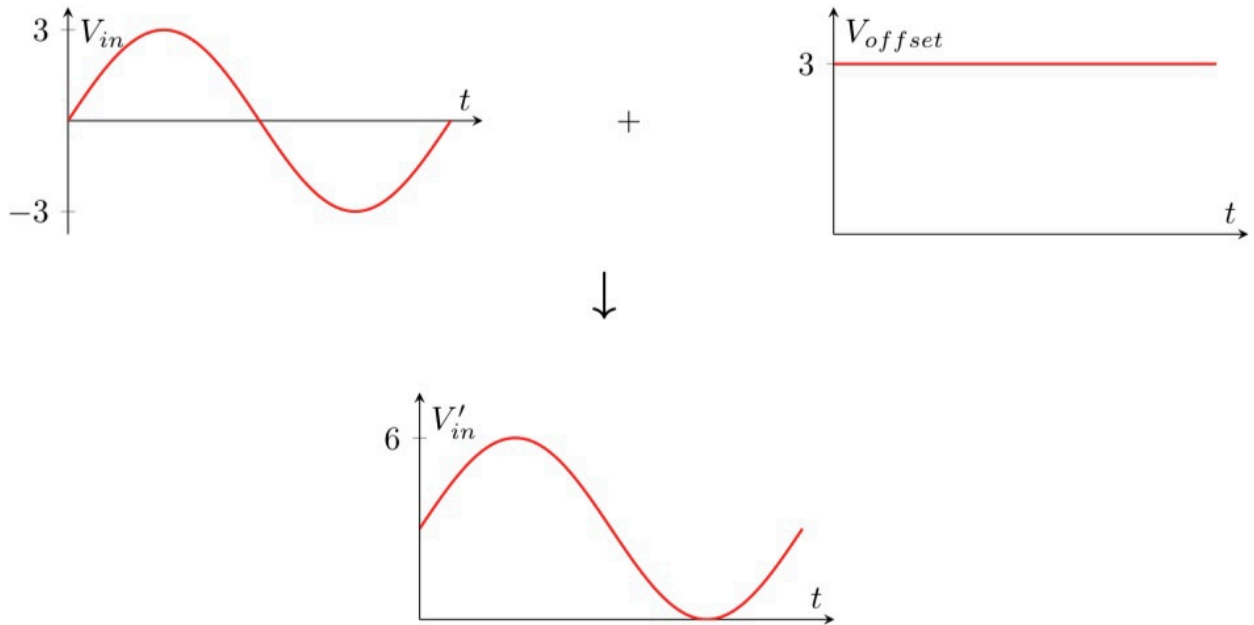


Figure 1.13: Adding offset to input signal

But the question is, how do we achieve this addition of offset? Well, it's easier than doing the ADC. Recall from CSE251, we studied inverting adders. If we simply apply the sinusoidal input to one input of an inverting adder, and a DC voltage of -3 V to another, we can get V'_{in} as in figure 1.13. Thus the full circuit including this adder would be:

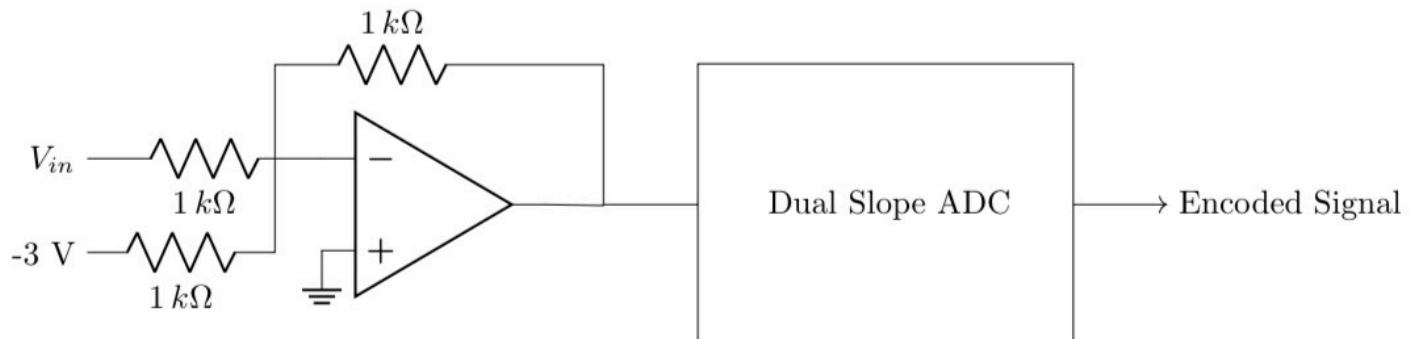


Figure 1.14: Modified Dual Slope ADC for handling bipolar input

Notice that we do not need to invert the input sinusoidal signal here, because it is bipolar. Two important jobs are still to be done. The reference voltage of the Dual Slope ADC also needs to be changed. As the input signal ranges between 0 V to 6 V now, the new reference voltage $V'_{ref} = 6$ V.

Also, we know that the minimum sampling frequency (Nyquist rate) must be twice of the input signal. So, we need to make sure that the maximum sampling rate of the Dual Slope ADC be at least 2 kHz. Thus, we need to modify the clock frequency.

$$\text{Worst conversion time, } t = 2 \times t_1 = 2 \times \frac{256}{f'_{CLK}}$$

$$\therefore f'_{CLK} = \frac{512}{t} = \frac{512}{\frac{1}{f_{Nyquist}}} = \frac{512}{\frac{1}{2 \times 10^3}} = 1.024 \text{ MHz}$$

Problem 5:

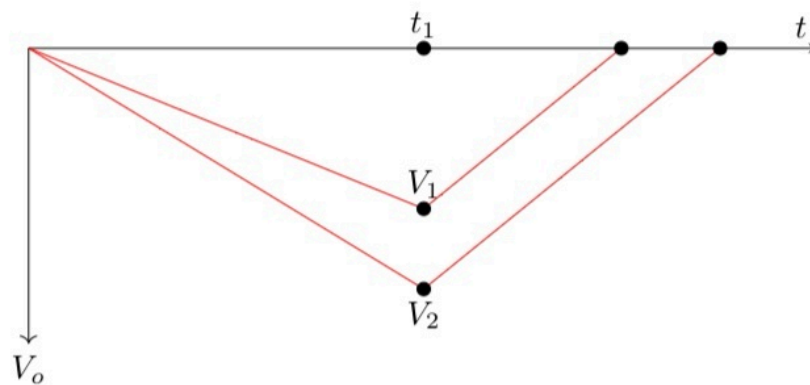


Figure 1.15: Integrator output vs time

For a 6-bit Dual Slope ADC $V_1 = -6$ V, $V_2 = -8$ V in figure 1.15 and the maximum allowable sampling frequency is 1.5625 kHz. The time constant τ of the integrator is 0.5 ms.

(a)	Find the two input voltages.
(b)	If the reference voltage is 15 V, find the digital outputs for the inputs found in (a).
(c)	Find the encoded output for which the slope of the integration phase of the integrator would be the average of the two given.

Solution:

(a)

$$V_{in1} = V_1 \times \frac{RC}{t_1} = -(-6) \times \frac{0.5 \times 10^{-3}}{t_1}$$

Now, worst case conversion time, $2t_1 = \frac{1}{1.5625 \times 10^3} = 0.64 \text{ ms}$, $\therefore t_1 = 0.32 \text{ ms}$.

$$\therefore V_{in1} = \frac{3 \times 10^{-3}}{0.32 \times 10^{-3}} = 9.375 \text{ V}$$

Similarly,

$$V_{in_2} = \frac{-(-8) \times 0.5 \times 10^{-3}}{0.32 \times 10^{-3}} = 12.5V$$

(b)

$$N_1 + 1 = \frac{V_{in_1}}{V_{ref}} \times 2^6 = \frac{9.375}{15} \times 64 = 40 \Rightarrow N_1 = 39$$

$$N_2 + 1 = \frac{V_{in_2}}{V_{ref}} \times 2^6 = \frac{12.5}{15} \times 64 = 53(\text{floored value}) \Rightarrow N_2 = 52$$

$$\therefore D_1 = \text{binary of } N_1 = 100111$$

$$\therefore D_2 = \text{binary of } N_2 = 110100$$

(c) Let slopes of integration for V_{in_1} & V_{in_2} are m_1 & m_2 respectively.

$$m_1 = -\frac{V_{in_1}}{RC} = -\frac{9.375}{0.5} = -18.75 V/ms$$

$$m_2 = -\frac{V_{in_2}}{RC} = -\frac{12.5}{0.5} = -25 V/ms$$

$$\therefore m = \frac{m_1 + m_2}{2} = -21.875 V/ms \Rightarrow -\frac{V_{in}}{RC} = -21.875 \Rightarrow V_{in} = 0.5 \times 21.875 = 10.9375V$$

$$\therefore N + 1 = \frac{V_{in}}{V_{ref}} \times 2^6 = \frac{10.9375}{15} \times 64 = 46(\text{floored value}) \Rightarrow N = 45$$

$$\therefore D = \text{binary of } 45 = 101101$$

Problem 6:

If for the 3-bit Flash ADC shown in figure 1.6, $V_2 = -1$ V and $V_6 = 7$ V—

(a)	Find the reference voltages V_{ref}^+ & V_{ref}^- .
(b)	What value of offset needs to be added to the input signal, for which the Flash ADC in (a) is designed, so that it can be implemented using an 8-bit Dual Slope ADC with a 2.5 MHz clock? What will be the reference voltage of the Dual Slope ADC?
(c)	Find the average conversion time of the Dual Slope ADC mentioned in (b), for the highest quantization interval in (a).

Solution:

(a) We see in figure 1.6, $V_2 = V_{ref}^- + I \times 1.5R$. Also, $V_6 = V_{ref}^- + I \times 5.5R$.

$$V_{ref}^- + 1.5 \times IR = -1$$

$$V_{ref}^- + 5.5 \times IR = 7$$

Solving the above two equations, $V_{ref}^- = -4V$, $IR = 2V$.

$$V_{ref}^+ = V_{ref}^- + 0.5IR + 6IR + 1.5IR = -4 + 8IR = -4 + 8 \times 2 = 12V$$

(b) $V_{offset} = 4V$.

$$V_{ref} = (V_{ref}^+ + V_{offset}) = (12 + -4) = 16V$$

(c) Highest quantization interval in (a): 9 V - 12 V.

Total conversion time for input of 9 V,

$$t_{9V} = \frac{2^n}{f_{CLK}} + \frac{9 + V_{offset}}{V_{ref}} \times \frac{2^n}{f_{CLK}} = \frac{2^8}{2.5 \times 10^6} \left(1 + \frac{9 + 4}{16}\right) = 0.1856 ms$$

Total conversion time for input of 12 V,

$$t_{12V} = \frac{2^n}{f_{CLK}} + \frac{12 + V_{offset}}{V_{ref}} \times \frac{2^n}{f_{CLK}} = \frac{2^8}{2.5 \times 10^6} \left(1 + \frac{12 + 4}{16}\right) = 0.2048 ms$$

Thus, the average conversion time for this quantization interval, $t_{avg} = \frac{0.1856 + 0.2048}{2} = 0.1952 ms$

Problem 7: For an 8-bit Dual Slope ADC, the maximum conversion time was found to be $512 \mu s$.

(a)	Find the clock frequency.
(b)	Find the maximum allowable sampling frequency.
(c)	Find the conversion time for the output '10110110'.

Solution:

(a) We have,

$$2t_1 = 512 \mu s \Rightarrow 2 \frac{2^n}{f_{CLK}} = 512 \mu s \Rightarrow f_{CLK} = 1 MHz$$

(b)

$$f_{s_{max}} = \frac{1}{2t_1} = 1953.125 Hz$$

(c)

$$t_{10110110} = t_1 + t_{2_{10110110}} = 256 + \frac{N+1}{2^n} t_1 = 256 + \frac{182+1}{256} 256 = 439 \mu s$$

Problems Roster

Problem 1:

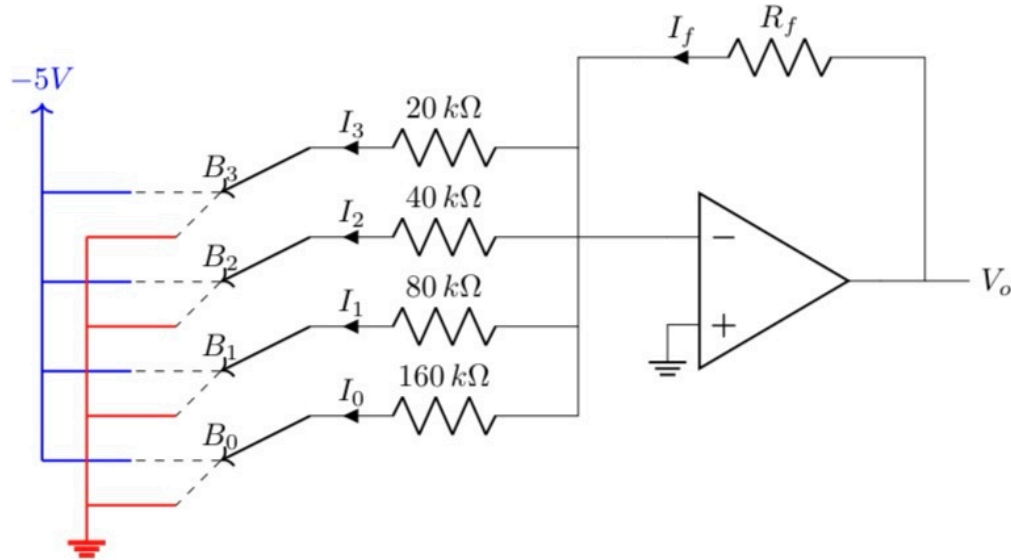


Figure 2.8: A 4-bit Binary Weighted Resistor DAC

For the 4-bit DAC circuit shown in figure 2.8, assume $R_f = 5\text{ k}\Omega$

(a)	Determine the output voltage for the inputs— (i) 1100 (ii) 1010.
(b)	Find the <i>MSB</i> & <i>LSB</i> voltages. What is the maximum possible output from the DAC?
(c)	What are the values of the input branch currents when they conduct?

Solution:

(a)

(i)

$$V_{o_{1100}} = -\frac{R_f}{R}(-V_{ref}^+ + \frac{-V_{ref}^+}{2} + \frac{V_{ref}^-}{4} + \frac{V_{ref}^-}{8}) = -\frac{5}{20}(-5 + \frac{-5}{2} + \frac{0}{4} + \frac{0}{8}) = 1.875\text{ V}$$

(ii)

$$V_{o_{1010}} = -\frac{5}{20}(-5 + 0 + \frac{-5}{4} + 0) = 1.5625\text{ V}$$

(b)

$$V_{MSB} = V_{o_{1000}} = -\frac{5}{20}(-5 + 0 + 0 + 0) = 1.25\text{ V}$$

$$V_{LSB} = V_{o_{0001}} = -\frac{5}{20}(0 + 0 + 0 + \frac{-5}{8}) = 0.15625\text{ V}$$

(c) The input branches conduct whenever their respective bits are "1". Thus,

$$I_3 = \frac{0 - (-5)}{20} = 0.25 \text{ mA}$$

$$I_2 = \frac{0 - (-5)}{40} = 0.125 \text{ mA}$$

$$I_1 = \frac{0 - (-5)}{80} = 0.0625 \text{ mA}$$

$$I_0 = \frac{0 - (-5)}{160} = 0.03125 \text{ mA}$$

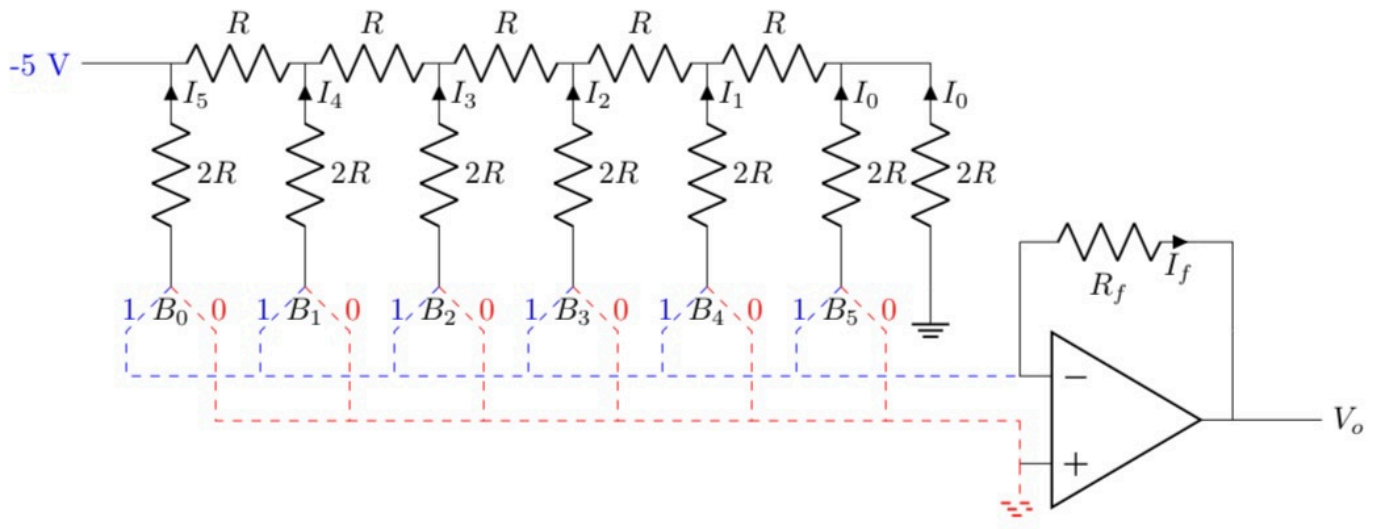
Problem 2:

Consider a 6-bit R2R DAC (variant 2). Assume $R_f = R = 5 \text{ k}\Omega$ & $|V_{ref}| = 5 \text{ V}$.

(a)	Draw the DAC circuit with proper labelings.
(b)	Find the input branch currents I_0 to I_5 .
(c)	What is the output for the input "010011"? What is the change in output voltage if the input changes from "101010" to "010101".

Solution:

(a)



(b)

$$I_5 = \frac{0 - (-5)}{2 \times 5} = 0.5 \text{ mA}$$

$$I_4 = \frac{1}{2}I_5 = 0.25 \text{ mA}$$

$$I_3 = \frac{1}{2}I_4 = 0.125 \text{ mA}$$

$$I_2 = \frac{1}{2}I_3 = 0.0625 \text{ mA}$$

$$I_1 = \frac{1}{2}I_2 = 0.03125 \text{ mA}$$

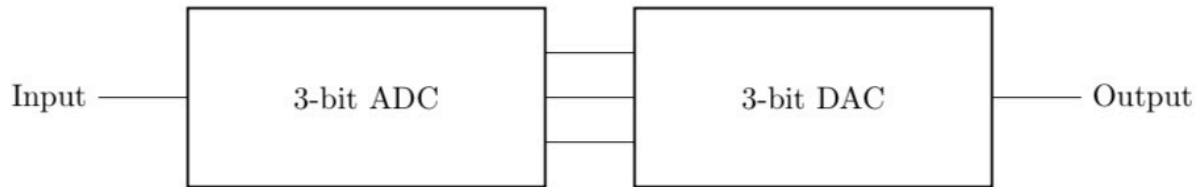
$$I_0 = \frac{1}{2}I_1 = 0.015625 \text{ mA}$$

(c)

$$V_{o_{010011}} = \frac{R_f|V_{ref}|}{2R} \left(0 + \frac{1}{2} + 0 + 0 + \frac{1}{16} + \frac{1}{32}\right) = \frac{5 \times 5}{2 \times 5} \times \frac{19}{32} = 1.484375 \text{ V}$$

$$\Delta V_o = V_{o_{101010}} - V_{o_{010101}} = \frac{R_f|V_{ref}|}{2R} \left[(1+0+\frac{1}{4}+0+\frac{1}{16}+0) - (0+\frac{1}{2}+0+\frac{1}{8}+0+\frac{1}{32})\right] = \frac{5 \times 5}{2 \times 5} \times \frac{21}{32} = 1.640625 \text{ V}$$

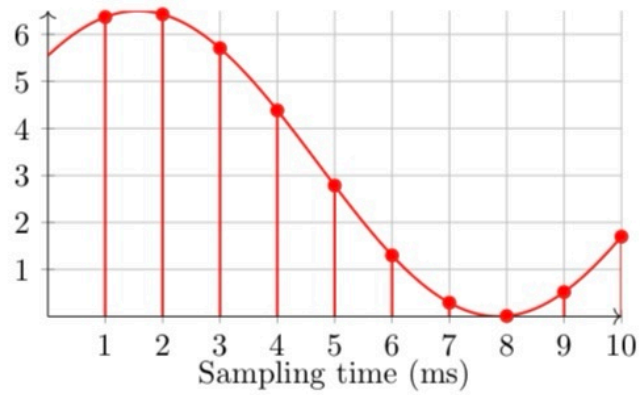
Problem 3:



ADC Input Voltage	Encoding	DAC Output Voltage
0-1	000	0.5
1-2	001	1.5
2-3	010	2.5
3-4	011	3.5
4-5	100	4.5
5-6	101	5.5
6-7	110	6.5
7-8	111	7.5

(a)	For the following input waveshape, draw the reconstructed output by the DAC. The sampling instances are marked on the input waveshape.
(b)	Comment on the quality of the reconstructed signal and how the quality would be affected if additional bits were used for both the ADC and DAC.

Input Voltage



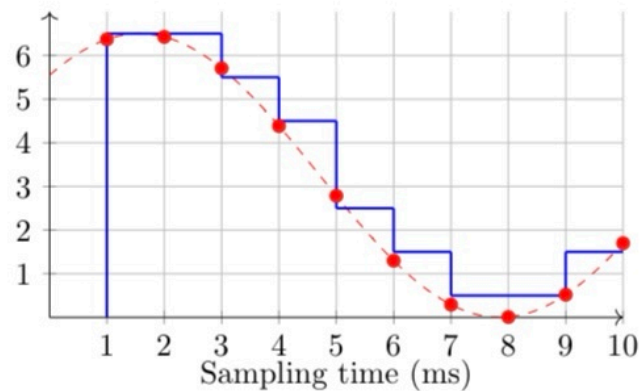
Solution:

(a) Using the table given, let's first find the output voltages for the sampled inputs.

Sampling Instances	Sampled Input	Encoding	DAC Output Voltage
1	6.3	110	6.5
2	6.4	110	6.5
3	5.7	101	5.5
4	4.4	100	4.5
5	2.8	010	2.5
6	1.3	001	1.5
7	0.3	000	0.5
8	0	000	0.5
9	0.5	000	0.5
10	1.7	001	1.5

Reconstructed Signal (in blue):

Output Voltage



(b) Reconstructed signal roughly estimates the original input, and the quality is not very good.

Problem 5:

Let a sender wishes to convert the analog signal $3.5 + 4\cos(\omega t)$ in a digital signal, and then transmit it to a receiver. The receiver will then back convert the received digital signal into an analog signal. But the receiver has a 3-bit R2R DAC, for which $R_f = R = 2\text{ k}\Omega$.

(a)	What should be the value of reference voltage for the receiver's DAC?
(b)	If the output from the DAC of the receiver at a certain instant is '3 V', what are the possible values of the input sampled value at the sender's end?
(c)	Calculate the step size of the DAC.

Solution:

(a) The minimum & maximum value of the input analog signal at the sender's end is $3.5 - 4 = -0.5$ & $3.5 + 4 = 7.5$ respectively. Considering mid-rise quantization, the quantization levels & encoded outputs for the ADC of this signal would be:

Quantization Levels	Encoded Outputs
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Thus, the highest quantization level is 7 V. This should be the output from the receiver's DAC for the input '111'. So, using the output voltage formula of R2R DAC,

$$V_{o_{111}} = \frac{R_f |V_{ref}|}{2R} \left(1 + \frac{1}{2} + \frac{1}{4}\right) \Rightarrow 7 = \frac{|V_{ref}|}{2} \times \frac{7}{4}$$
$$\Rightarrow |V_{ref}| = 8\text{ V}$$

(b) From the table in (a), we see, the quantization level 3 was encoded into '011'. Thus, this should be the input at the receiver's DAC for an output of 3 V.

(c)

$$\Delta = \frac{R_f |V_{ref}|}{2R \cdot 2^{n-1}} = \frac{1}{2} \frac{8}{2^{3-1}} = 1\text{ V}$$

BRAC UNIVERSITY
Department of Computer Science & Engineering
Practice Problem sheet (Week5)
CSE 350: Digital Electronics and Pulse Technique

Question 1

(a)	An analog signal in the range $-1V$ to $16V$ is to be converted to a digital signal with a quantization error of less than or equal to 0.781% of the input voltage range for quantization. e.g the overall quantization error for this input signal should be less than or equal to 0.781% of the ADC's maximum quantization error. The quantization error is the maximum error occurring after quantizing the analog signal. Find the required number of bits for the above expression.
(b)	Determine the minimum sampling frequency of a signal.

$$a) \Delta Q = \text{Quantization error} = \frac{1}{2} \frac{V_{\max} - V_{\min}}{2^N}$$

$$\% \text{ Quantization error} = \frac{0.781}{100} \times (V_{\max} - V_{\min})$$

$$\Rightarrow \frac{\Delta Q}{V_{\max} - V_{\min}} = \frac{0.781}{100}$$

$$\Rightarrow \frac{1}{2} \times \frac{V_{\max} - V_{\min}}{2^N} \times \frac{1}{V_{\max} - V_{\min}} = \frac{0.781}{100}$$

$$\Rightarrow \frac{1}{2^{N+1}} = \frac{0.781}{100}$$

$$\Rightarrow 2^{N+1} = 128 \Rightarrow \boxed{N = 6}$$

$$b) \boxed{f_{\text{sampling}} = 2 \times f_{\text{input}}}$$

Problems Roster

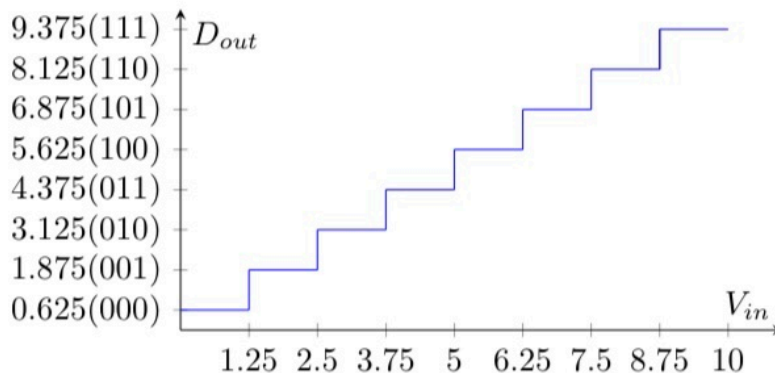
Problem 1: Design a 3-bit Flash ADC for a signal with maximum & minimum value of 10 V & 0 V respectively. The quantized levels of the ADC should be uniform (equal step-size). Answer the following:

(a)	Calculate the total number of resistors & Op Amps required.
(b)	Calculate the 1LSB value or step size.
(c)	Find the quantization levels & plot the D_{out} (digital output) vs V_{in} plot.
(d)	If the input is 7 V: (i) Find the quantization interval the input lies in. (ii) What is the digital output for this input?
(e)	If the digital output is 110_2 , what are the maximum & minimum values of input that would produce this result?
(f)	If an additional bit was desired at the output, what would be the required number of components?

Solution:

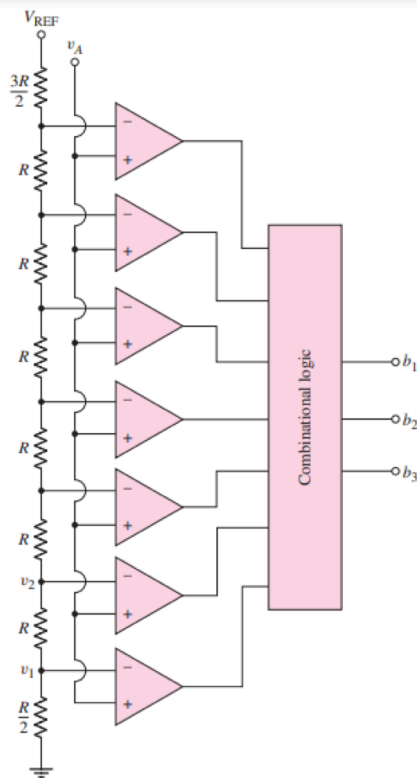
- (a) $N_R = 2^n = 2^3 = 8$
 $N_{OA} = 2^n$ or $2^n - 1 = 8$ or 7
 (b) $V_{1LSB} = \Delta = \frac{V_{max} - V_{min}}{2^n} = \frac{10 - 0}{8} = 1.25 \text{ V}$
 (c) The quantization levels are as follows:

Quantization Interval	Quantization Level
0-1.25	0.625
1.25-2.5	1.875
2.5-3.75	3.125
3.75-5	4.375
5-6.25	5.625
6.25-7.5	6.875
7.5-8.75	8.125
8.75-10	9.375



Design a 3 bit flash ADC with a reference voltage of $V_{REF}=10V$. The quantized levels for this ADC should be **uniform (Equal step size)**.

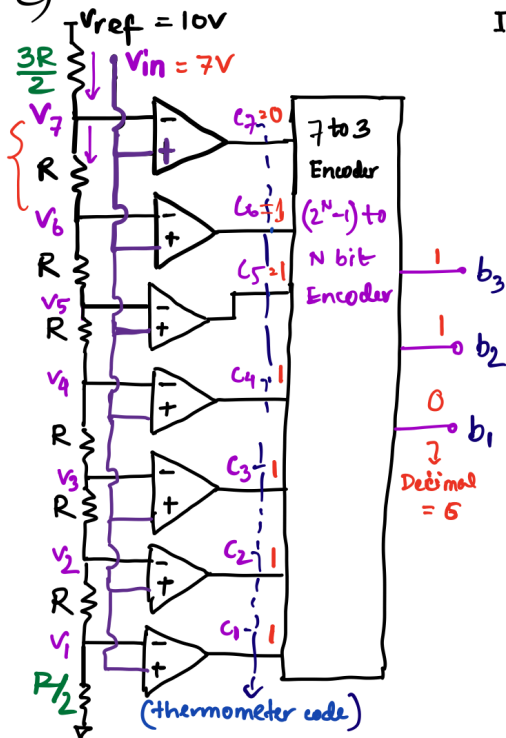
(a)	Calculate total number of resistors and comparators required to fabricate a 3-bit flash A/D converter.
(b)	Calculate the 1LSB value or each step size (quantization range) for this 3 bit ADC. What would be the resolution for this ADC?
(c)	Calculate each quantization level and plot the Vin vs Dout graph.
(d)	If the input voltage Vin= 7 V , i) Comment on the quantization range in which the input lies. ii) Find the digital output for the given input.
(e)	If the 3-bit output is 110 What is the maximum and minimum value of v_A that produces this output?(Hint:find the quantization range).
(f)	If 1 bit was increased from this 3 bit Flash ADC, how many resistors and comparators are required? comment on the hardware changes for this 1 bit of increment.



a) $N = 3$; Resistor required $= 2^N = 2^3 = 8$
 Comparator required $= 2^3 - 1 = 7$

b) 1 LSB = resolution or step size = $\frac{V_{max} - V_{min}}{2^N} = \frac{10 - 0}{2^3} = 1.25V$

c)



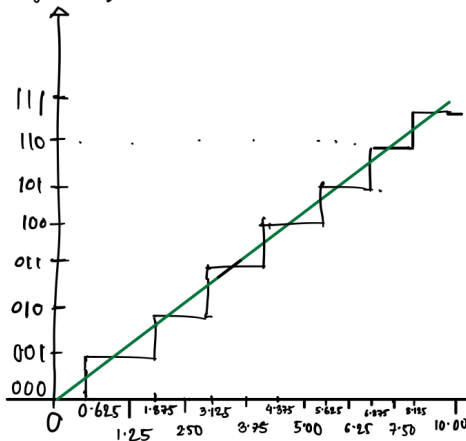
$$I = \frac{10V}{\frac{3R}{2} + 6R + \frac{R}{2}} = \frac{10V}{R(\frac{3}{2} + \frac{12}{2} + \frac{1}{2})} = \frac{10V}{R \cdot \frac{16}{2}} = \frac{10}{8R} = \frac{1.25}{R}$$

(same as previous example)

$V_0 = 0V$
 $V_1 = \frac{1.25}{R} \cdot R/2 = 0.625V (\frac{1}{2} \text{ LSB})$
 $V_2 = \frac{1.25}{R} \cdot (R + \frac{R}{2}) = \frac{1.25}{R} \cdot \frac{3R}{2} = 1.875V (\frac{3}{2} \text{ LSB})$
 $V_3 = \frac{1.25}{R} \cdot (\frac{5R}{2}) = 3.125V$
 $V_4 = \frac{1.25}{R} \cdot \frac{7R}{2} = 4.375V$
 $V_5 = \frac{1.25}{R} \cdot \frac{9R}{2} = 5.625V$
 $V_6 = \frac{1.25}{R} \cdot \frac{11R}{2} = 6.875V$
 $V_7 = \frac{1.25}{R} \cdot \frac{13R}{2} = 8.125V$
 $V_{ref} = \frac{1.25}{R} \cdot (\frac{15R}{2} + \frac{3R}{2}) = 10V$

input 7

Digital output code, D



Input V_{in} vs Digital output code, D

As all the step size is uniform or equal to 1LSB; this distribution is a uniform distribution. from $(V_1 - V_7)$

d) if input = 7V ; it will be in between the quantization range of $V_6 - V_7$ or 6.875V - 8.125V.

Quantization Range	Quantized value	Digital o/p ($b_2 b_1 b_0$)
6.875V - 8.125V	6	110

e) If digital output = 110 = 6 (quantized value)

Maximum value of $V_A = 8.125V$

Minimum value of $V_A = 6.875V$

f) If 1 bit was increased $\rightarrow N = 3 + 1 = 4$ bit

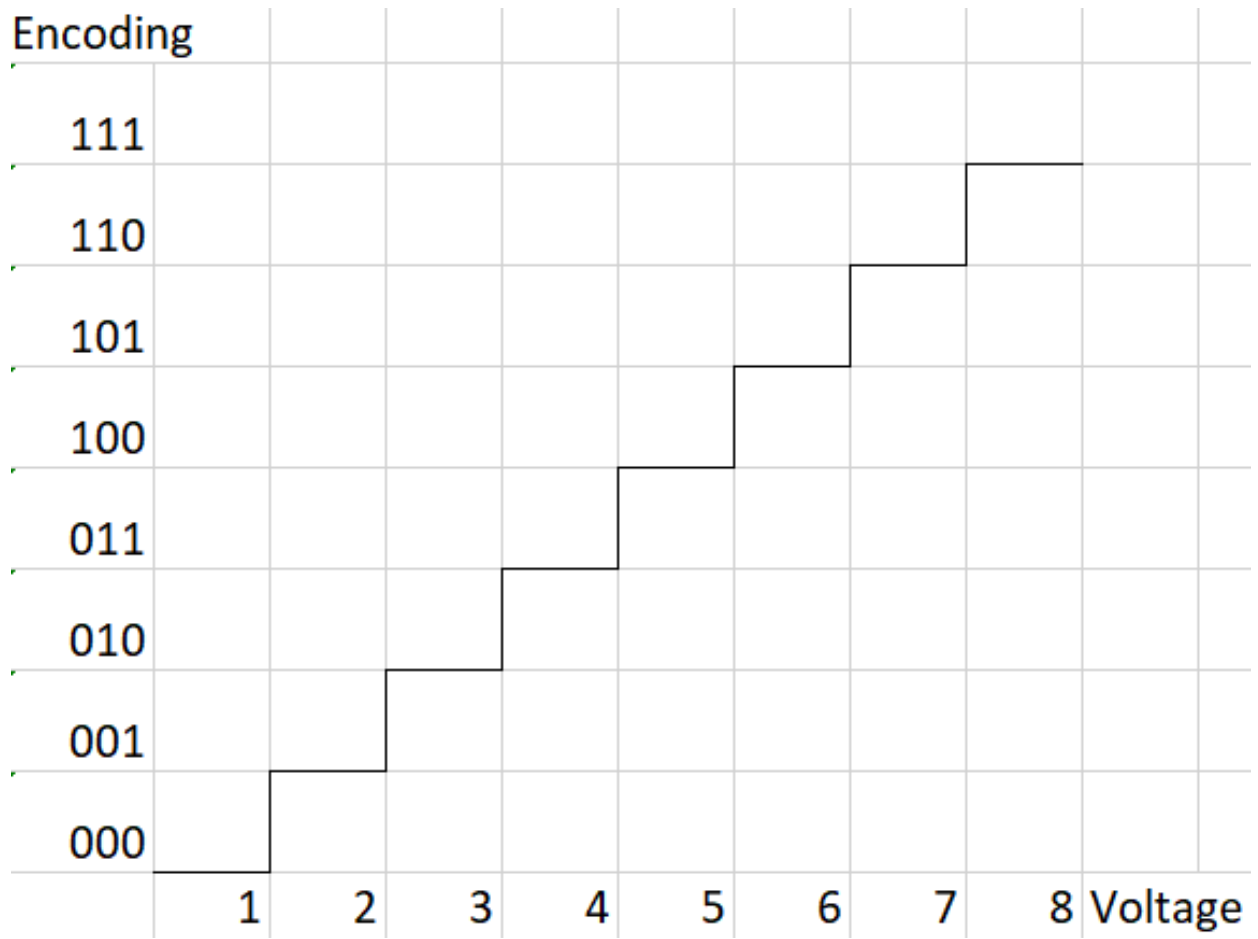
No. of R = $2^N = 2^4 = 16$

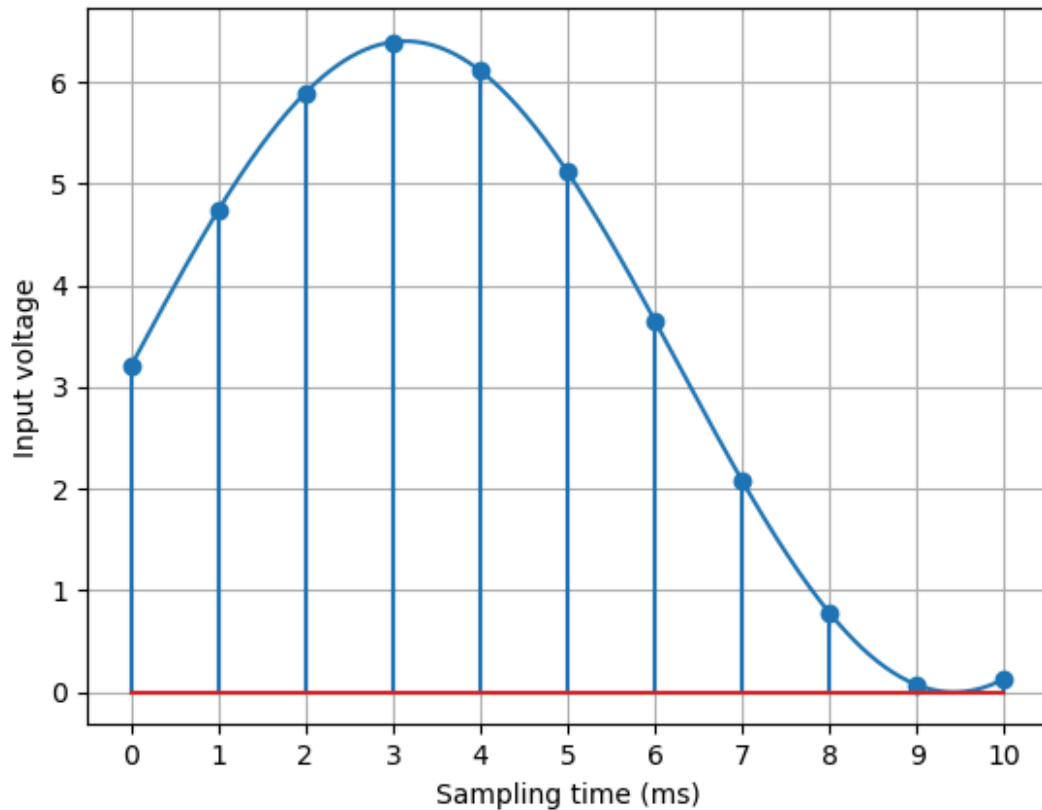
No. of Comparator = $2^4 - 1 = 15$

Encoder = 15 lines to 4 lines

1 bit of increment doubles the hardware needed.

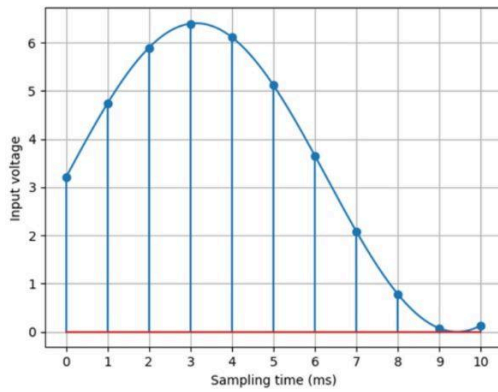
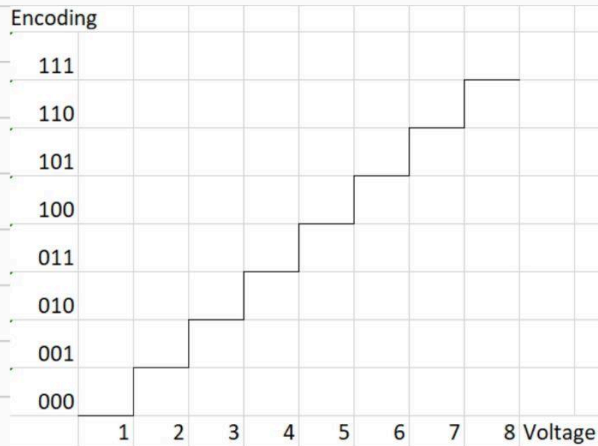
Question 3



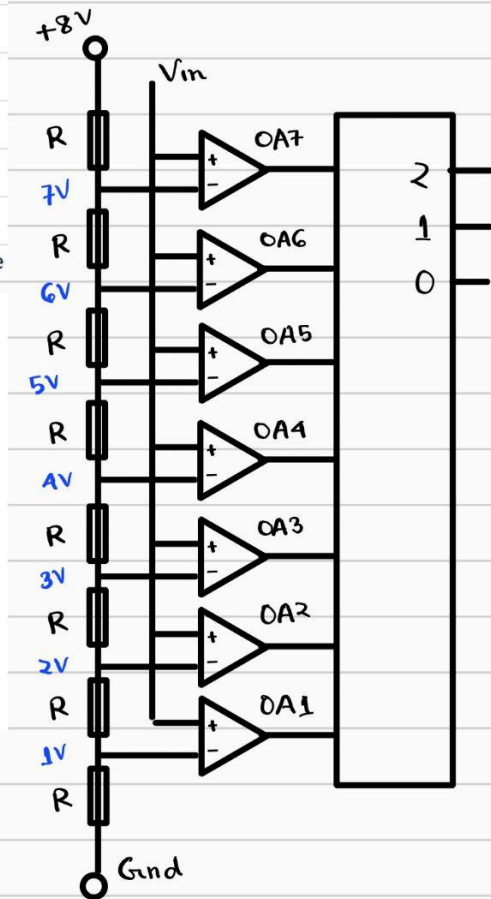


(a)	Design a 3-bit Flash ADC circuit for the given input-output characteristics.
(b)	An unknown signal is passed to the ADC as input. The ADC takes samples at a rate of 1kHz. In the graph, the input signal is shown where the x-axis represents time, and the y-axis represents the voltage at any specific time. The sampling instances are given in the figure. Find the encoded output that represents the analog signal inside the given time frame.

Solution:



③ 3 bit ADC - $2^3 - 1 = 7$
Op amps and 8 resistors
required.



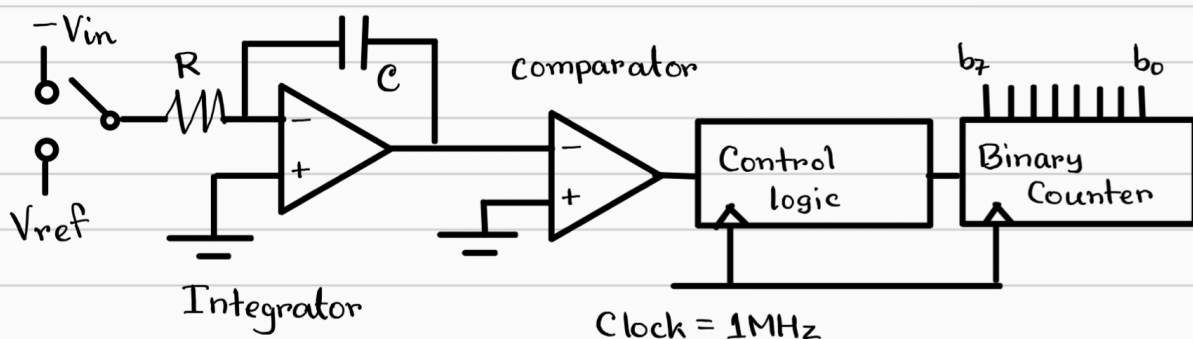
⑥

Sampling instance	Level	Encoding
0	3	011
1	4	100
2	5	101
3	6	110
4	6	110
5	5	101
6	3	011
7	2	010
8	0	000
9	0	000
10	0	000

Question 4

Dual-Slope A/D Converter

- A dual slope A/D converter has $V_{ref} = 5V$, and a 8 bit counter that outputs an 8-bit representation of the input signal. A 1 MHz clock is used for the clock and counter circuits



- ① Determine the input voltage range, number of steps and step size (resolution) of the A/D converter.
- ② For a specific input voltage, the ADC outputs a count of $m=100$. Determine the input voltage.
- ③ Determine the total time required to get the reading in ②
- ④ Calculate the maximum sampling rate of the ADC.
- ⑤ A person wishes to get a sampling rate 4 times that of the current circuit. Determine how many bits should be used for the counter without changing any other system parameters.

① Input voltage range : $5V$ ($0 - V_{ref}$)
 number of steps $= 2^n = 2^8 = 256$
 step size $= \frac{\text{voltage range}}{\text{number of steps}} = 0.0195 V$

② We know, $V_{in} = \frac{m}{2^n} V_{ref} = \frac{100}{256} \times 5V = 1.95V$

③ For a single conversion, the integrator

Ⓐ integrates $-V_{in}$ for t_1 duration as counter counts from $0 - 2^n$

$$t_1 = \frac{2^n}{\text{clock rate}} = \frac{256}{1 \times 10^6} = 2.56 \times 10^{-4} s$$

Ⓑ integrates V_{ref} for t_2 duration as counter counts from $0 - m$

$$t_2 = \frac{m}{\text{clock rate}} = \frac{100}{1 \times 10^6} = 1 \times 10^{-4} s$$

$$\text{total conversion time} = t_1 + t_2 = 3.56 \times 10^{-4} s$$

④ Maximum sampling rate is limited by the time required for worst case conversion of input $= 5V$

$$\text{In that case, } t_1 = t_2 = 2.56 \times 10^{-4} s$$

$$\text{thus, worst case conversion duration} = t_1 + t_2 = 5.12 \times 10^{-4} s$$

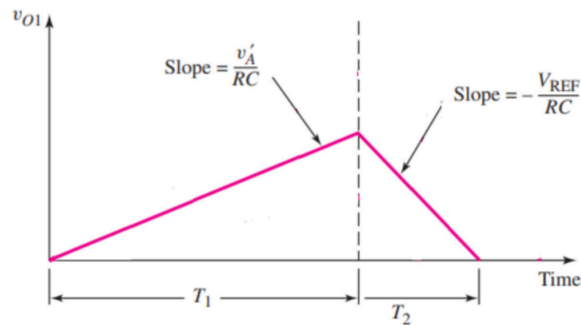
$$\text{Max sampling rate} = \frac{1}{5.12 \times 10^{-4} s} = 1953.125 \text{ Hz}$$

⑤ Sampling duration $\propto 2^n$ // Sampling rate $\propto \frac{1}{2^n}$

$$\text{For 4 times sampling rate, counter bit} = (8 - 2) = 6$$

$$f) \frac{D}{2^N} = \frac{V_A'}{V_{REF}} \Rightarrow D = \frac{V_A' \times 2^N}{V_{REF}} = \frac{3}{5} \times 2^8 = 153.6 \approx 153$$

Digital o/p (binary value of 153) = 10011001



(a)	Find the value of v_{O1} at $t=T_1$, suppose $R=25k\Omega$ and $C=2\mu F$.
(b)	If $R=50k\Omega$ and we $C=2\mu F$, calculate the new value of V_{O1} at $t=T_1$, compare it with (a) and comment on the changes between two output voltages.

$$a) \text{ slope} = \frac{\Delta Y}{\Delta x} = \tan \theta = \frac{OB}{AB}$$

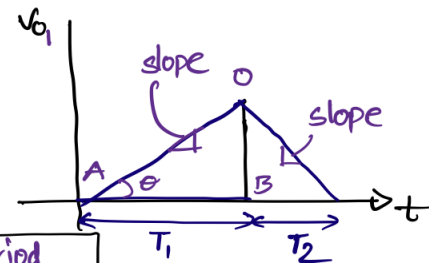
$$\text{As, } \tan \theta = \frac{V_A'}{RC}; \text{ so } \frac{OB}{AB} = \frac{V_A'}{RC}$$

$$\Rightarrow v_{O1} = OB = \frac{V_A'}{RC} \times AB$$

$$\Rightarrow v_{O1} = \frac{V_A'}{RC} \times T_1$$

$$\Rightarrow v_{O1} = \frac{2}{25K \times 2\mu F} \times 2^6 \times 6\mu S$$

$$v_{O1} = 0.01536V$$



$$T_1 = 2^N \cdot 1 \text{ clock period} \\ = 2^N \cdot T = 2^6 \cdot 6\mu S$$

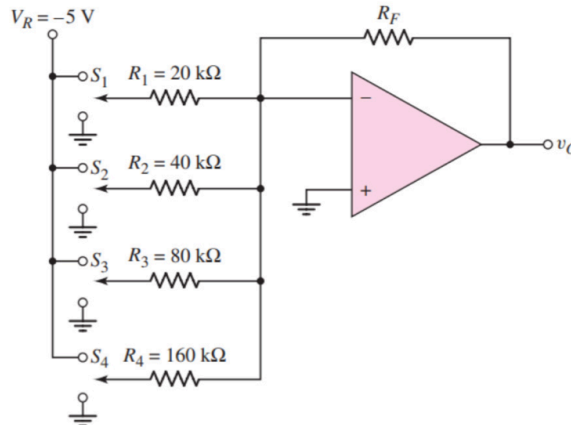
$$b) v_{O1} = \frac{2}{50K\Omega \times 2\mu F} \times 2^6 \times 6\mu S$$

$$v_{O1} = 0.00768V$$

Comment:

As, resistance increases by 2, it decreases the output by half.

(a)	Determine the output voltage, v_o in V of the 4-bit weighted-resistor D/A in the following figure for input = 1010 and input = 1100? Assume $R_F = 5k\Omega$.
(b)	Identify the maximum allowed tolerance ($\pm$ percent) in the value of R_1 so that the maximum error in the output is limited to $\pm \frac{1}{2}$ LSB quantized voltage value? (hint: Do two separate calculations for output error being $+\frac{1}{2}$ LSB and $-\frac{1}{2}$ LSB. Then, calculate the percentage change in R_1 in these two cases.)



a) Input = 1010; $R_F = 5k\Omega$

$$V_o = 5 \times \frac{5}{20} + 5 \times \frac{5}{80} = \boxed{1.5625V}$$

Input = 1100; $R_F = 5k\Omega$

$$V_o = 5 \times \frac{5}{20} + 5 \times \frac{5}{40} = \boxed{1.875V}$$

b) $1 \text{ LSB} = 5 \times \frac{5}{160} = 0.15625V$

$$\Rightarrow \frac{1}{2} \text{ LSB} = 0.078125V$$

$$1 \text{ MSB} = 1.25V \rightarrow$$

$$V_o = 1.25 \pm 0.078125V$$

$$= 1.328125V \text{ or } 1.171875V$$

$$-5 \times \frac{5}{R_1} = 1.328125V$$

or

$$-5 \times \frac{5}{R_1} = 1.171875V$$

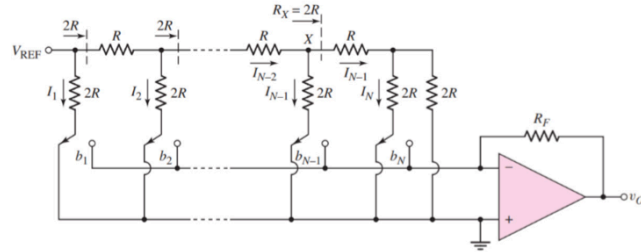
$\Rightarrow$

$$R_1 = 18.823k\Omega \rightarrow \% \text{ change in } R_1 = \frac{20 - 18.823}{20} \times 100 = \boxed{5.88\%}$$

$$\text{or } 21.33k\Omega \rightarrow \% \text{ change in } R_1 = \frac{21.33 - 20}{20} \times 100 = 6.67\%$$

$$\boxed{\text{Max tolerance} \rightarrow 5.88\%}$$

$$\left[\begin{array}{l} V_o = 1 \text{ MSB} + \frac{1}{2} \text{ LSB} \\ \text{or} \\ V_o = 1 \text{ MSB} - \frac{1}{2} \text{ LSB} \end{array} \right] \text{ Output error}$$



The N -bit D/A converter with an R - $2R$ ladder network in the above figure is to be designed as a 6-bit D/A device. Suppose $V_{REF} = -5V$ and $R_F = R = 5k\Omega$.

(a)	Find currents $I_1, I_2, I_3, I_4, I_5, I_6$.
(b)	What is the output voltage if the input is 010011?
(c)	What is the change in output voltage if the input changes from 101010 to 010101?

$$a) \quad I_1 = \frac{V_{REF}}{2R} = \frac{-5}{10} \Rightarrow I_1 = -0.50 \text{ mA}$$

$$I_2 = \frac{I_1}{2} = -0.25 \text{ mA}$$

$$I_3 = \frac{I_2}{2} = -0.125 \text{ mA}$$

$$I_4 = \frac{I_3}{2} = -0.0625 \text{ mA}$$

$$I_5 = \frac{I_4}{2} = -0.03125 \text{ mA}$$

$$I_6 = \frac{I_5}{2} = -0.015625 \text{ mA}$$

$$b) \quad \text{Input} = 010011$$

$$V_o = -[I_2 + I_5 + I_6] R_F = [0.25 + 0.03125 + 0.015625] \text{ mA} \times 5k\Omega = 1.484375 \text{ V}$$

$$c) \quad \text{Input} = 101010$$

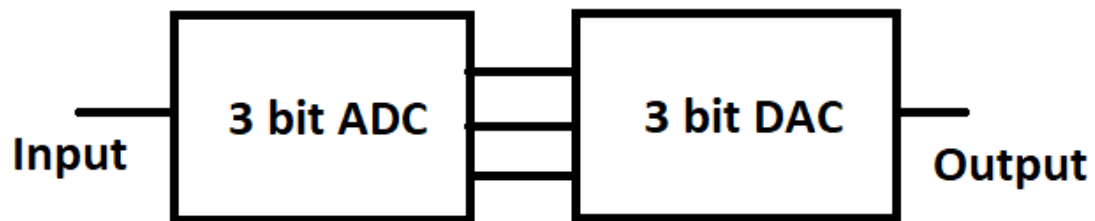
$$V_o = -[I_1 + I_3 + I_5] R_F = [0.50 + 0.125 + 0.03125] \text{ mA} \times 5k\Omega = 3.28125 \text{ V}$$

$$\text{Input} = 010101$$

$$V_o = -[I_2 + I_4 + I_6] R_F = [0.25 + 0.0625 + 0.015625] \text{ mA} \times 5k\Omega = 1.640625 \text{ V}$$

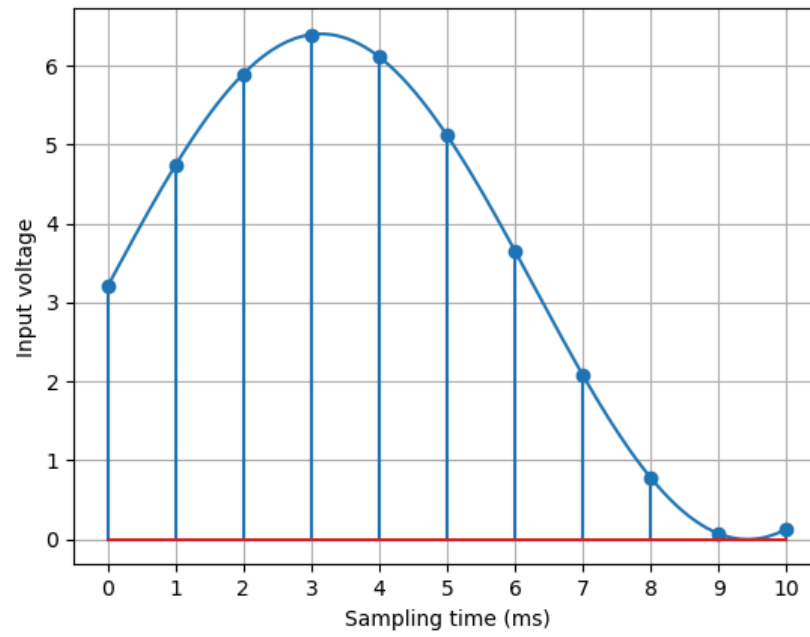
$$\Delta V_o = 1.640625 \text{ V}$$

Question 7



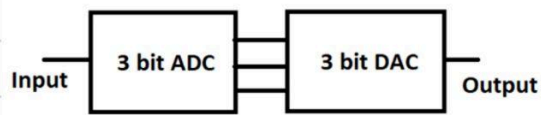
ADC Input Voltage	Encoding	DAC Output Voltage
0-1	000	0.5
1-2	001	1.5
2-3	010	2.5
3-4	011	3.5
4-5	100	4.5
5-6	101	5.5
6-7	110	6.5
7-8	111	7.5

(a)	For the following input waveshape, draw the reconstructed output by the DAC. The sampling instances are marked on the input waveshape.
(b)	Comment on the quality of the reconstructed signal and how the quality would be affected if additional bits were used for both the ADC and DAC.

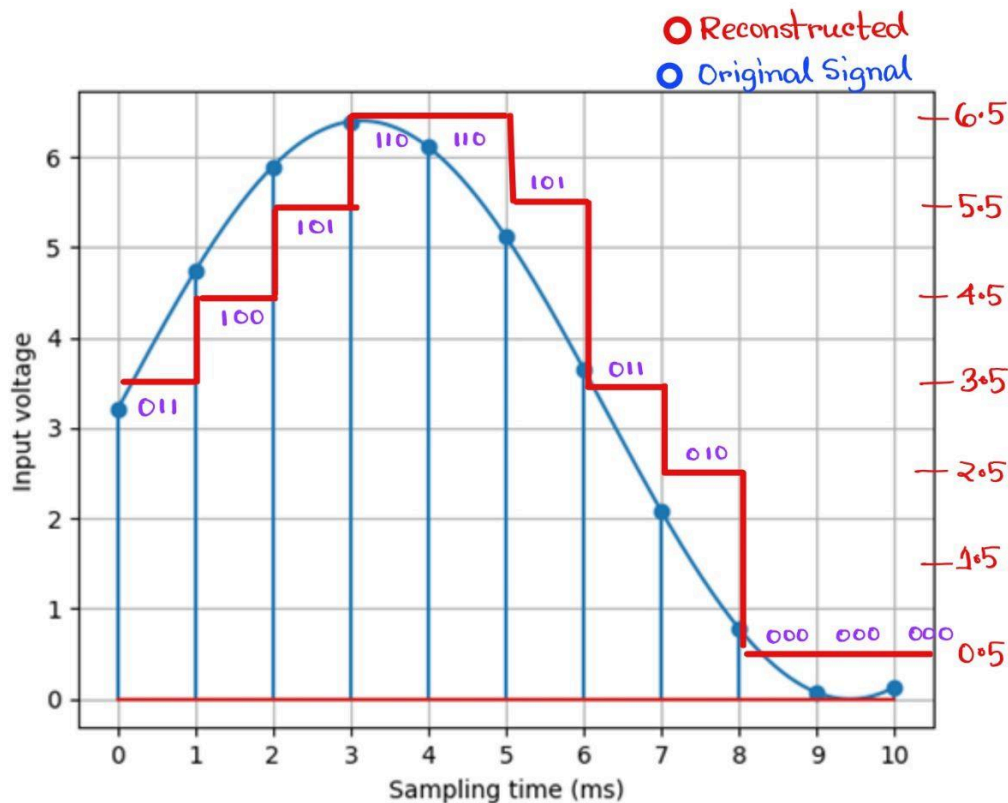


Solution:

a



ADC Input Voltage	Encoding	DAC Output Voltage
0-1	000	0.5
1-2	001	1.5
2-3	010	2.5
3-4	011	3.5
4-5	100	4.5
5-6	101	5.5
6-7	110	6.5
7-8	111	7.5



b

Reconstructed signal roughly estimates the original input, and the quality is not very good.

Using higher number of bits for both the ADC and DAC will give a better voltage resolution. The reconstructed voltage levels would be closer to that of the originally sampled value.